

Problem Solving

1. Let $X_1, \dots, X_n$ be a random sample from a Poisson distribution with mean $\theta > 0$.
- (a) Show that the likelihood ratio test of $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$ is based upon the statistic $Y = \sum_{i=1}^n X_i$. Obtain the null distribution of Y .
- (b) For $\theta_0 = 2$ and $n = 5$, find the significance level of the test that rejects H_0 if $Y \leq 4$ or $Y \geq 17$.

2. Let $X_1, \dots, X_n$ be a random sample from a Poisson distribution with mean $\theta > 0$. Test $H_0 : \theta = 2$ vs $H_1 : \theta \neq 2$ using

- (a) $-2 \log \Lambda$
- (b) a Wald-type statistic
- (c) Rao's score statistic

3. The Pareto distribution is a frequently used model in the study of income and has the distribution function $F(x; \theta_1, \theta_2) = 1 - (\theta_1/x)^{\theta_2}$ for $\theta_1 \leq x$, zero elsewhere, where $\theta_1 > 0$ and $\theta_2 > 0$. If $X_1, \dots, X_n$ is a random sample from this distribution, find the MLEs of θ_1 and θ_2 .

4. Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ be independent random samples from the two normal distributions $N(0, \theta_1)$ and $N(0, \theta_2)$. respectively.

- (a) Find the likelihood ratio Λ for testing $H_0 : \theta_1 = \theta_2$ against $H_1 : \theta_1 \neq \theta_2$.
- (b) This Λ is a function of what F - statistic that would actually be used in this test?

